Chemical Process Equipment Design (CHC306)

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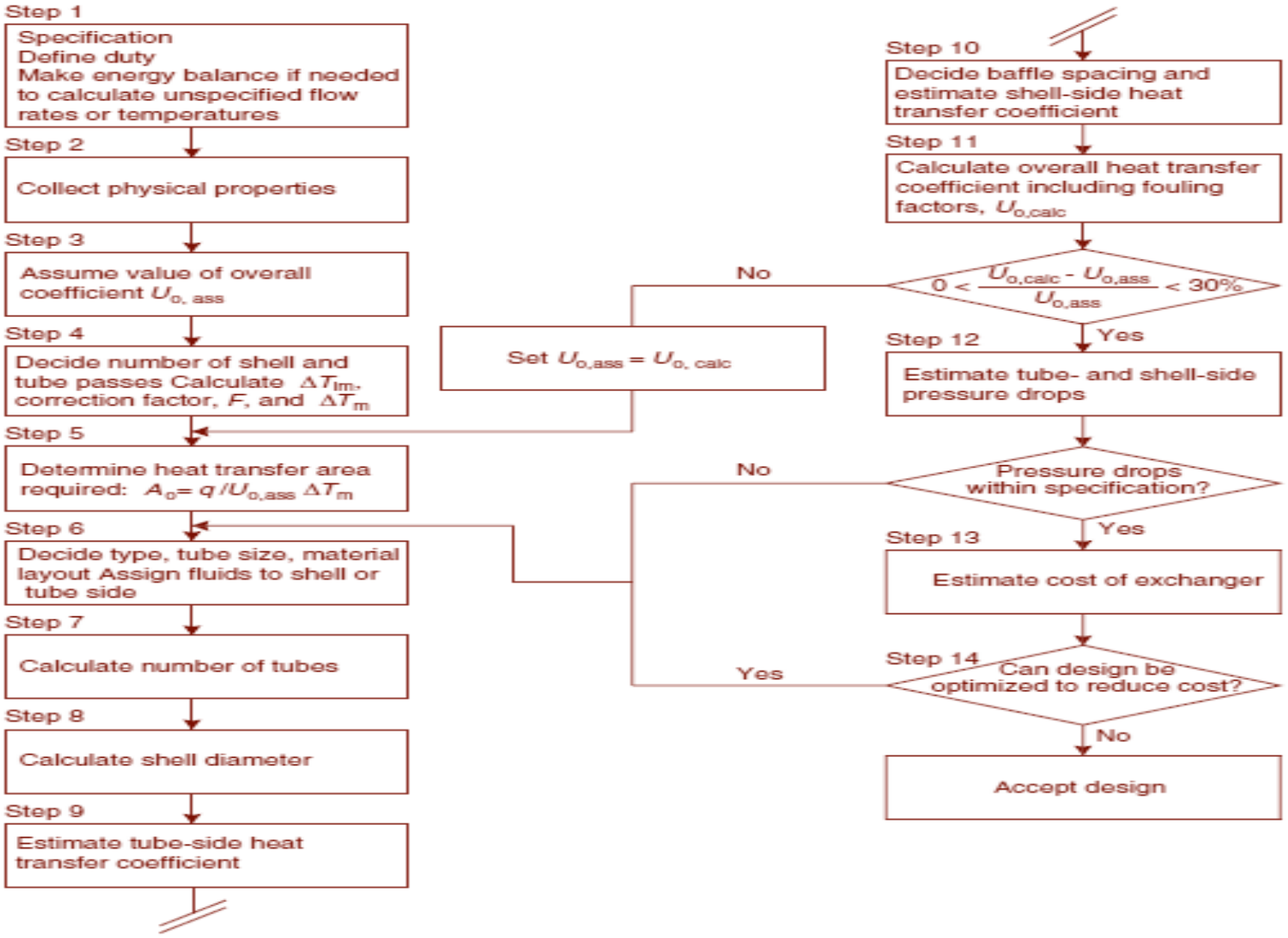
Module-2.2 : Design of heat transfer equipment

- There are two general classes of problems encountered by a process engineer. One class is the design problem, which requires calculating the size of a process unit. The other class is the rating problem, which requires determining if an existing process unit will satisfy process conditions.
- For a heat exchanger, the sizing problem is calculating surface area for transferring a specified amount of heat.
- For the rating problem, the heat-transfer area is fixed.
- The prime objective in the design of an exchanger is to determine the surface area required for the specified duty (rate of heat transfer) using the temperature differences available.
- As the physical layout of the exchanger cannot be determined until the area is known the design of an exchanger is of necessity a trial and error procedure.

The steps in a typical design procedure are given below:

1. Define the duty: heat-transfer rate, fluid flow-rates, temperatures.
2. Collect together the fluid physical properties required: density, viscosity, thermal conductivity.
3. Decide on the type of exchanger to be used.
4. Select a trial value for the overall coefficient, U.
5. Calculate the mean temperature difference, ΔT_m.
6. Calculate the area required.
7. Decide the exchanger layout.
8. Calculate the individual coefficients.
9. Calculate the overall coefficient and compare with the trial value. If the calculated value differs significantly from the estimated value, substitute the calculated for the estimated value and return to step 6.
10. Calculate the exchanger pressure drop; if unsatisfactory return to steps 7 or 4 or 3, in that order of preference.
11. Optimise the design: repeat steps 4 to 10, as necessary, to determine the cheapest exchanger that will satisfy the duty. Usually this will be the one with the smallest area.

Design procedure for shell-and-tube heat exchangers



- **Overall heat-transfer coefficient:**

The general equation for heat transfer across a surface is:

$$Q = UA\Delta T_m$$

where Q = heat transferred per unit time, W,
U = the overall heat transfer coefficient, $W/m^2{}^\circ C$,
A = heat-transfer area, m^2,
ΔT_m = the mean temperature difference, the temperature driving force, $^\circ C$.

- The overall coefficient is the reciprocal of the overall resistance to heat transfer, which is the sum of several individual resistances.

$$\frac{1}{U_o} = \frac{1}{h_o} + \frac{1}{h_{od}} + \frac{d_o \ln\left(\frac{d_o}{d_i}\right)}{2k_w} + \frac{d_o}{d_i} \times \frac{1}{h_{id}} + \frac{d_o}{d_i} \times \frac{1}{h_i}$$

where U_o = the overall coefficient based on the outside area of the tube, $W/m^2{}^\circ C$,
h_o = outside fluid film coefficient, $W/m^2{}^\circ C$,
h_i = inside fluid film coefficient, $W/m^2{}^\circ C$,
h_{od} = outside dirt coefficient (fouling factor), $W/m^2{}^\circ C$,
h_{id} = inside dirt coefficient, $W/m^2{}^\circ C$,
k_w = thermal conductivity of the tube wall material, $W/m^\circ C$,
d_i = tube inside diameter, m,
d_o = tube outside diameter, m.

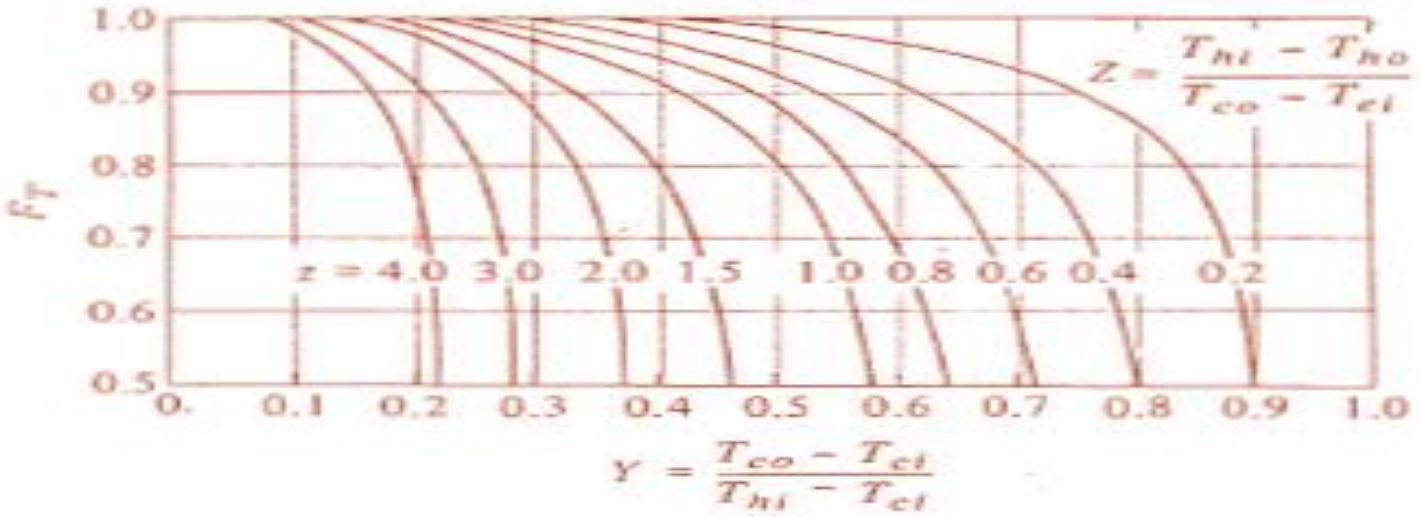
Multipass and Cross-Flow Heat Exchangers: Use of a Correction Factor

The log mean temperature difference relation developed earlier is limited to parallel-flow and counter-flow heat exchangers only.

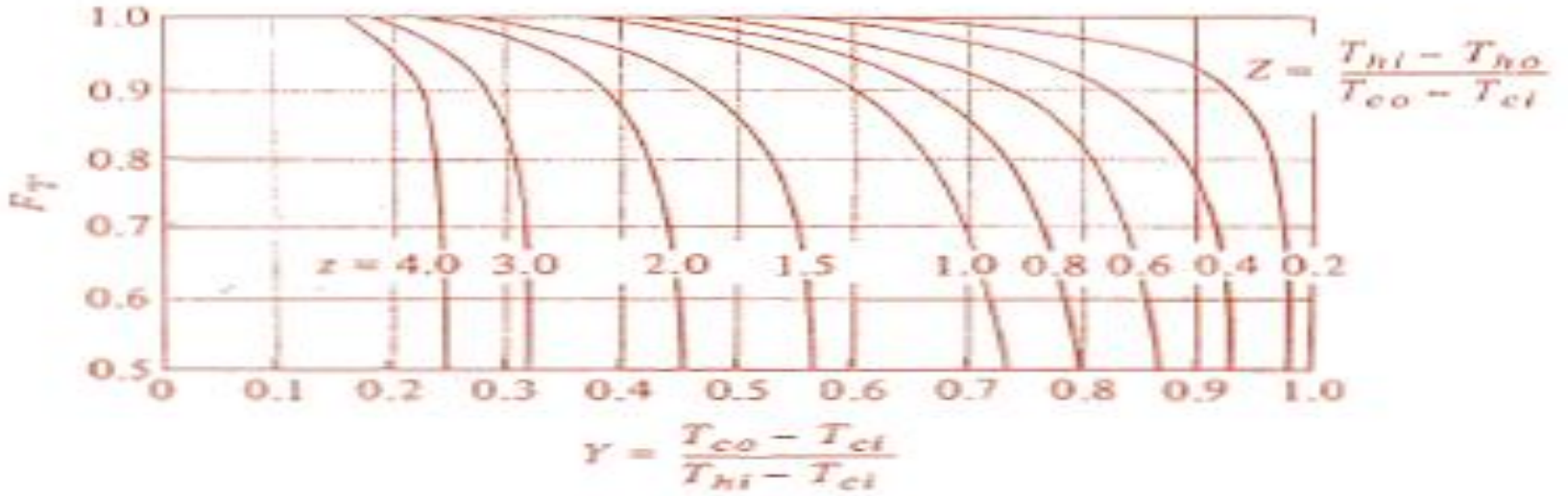
To simplify the analysis of *cross-flow* and *multipass shell-and-tube* heat exchangers, it is convenient to express the log mean temperature difference relation as

$$\Delta T_{lm} = F \Delta T_{lm,CF}$$

F is the **correction factor**, and $\Delta T_{\text{lm, CF}}$ is the log mean temperature for counter-flow case.



(a)



(b)

FIGURE 4.9-4. *Correction factor F_T to log mean temperature difference: (a) 1–2 exchangers, (b) 2–4 exchangers. [From R. A. Bowman, A. C. Mueller, and W. M. Nagle, Trans. A.S.M.E., 62, 284, 285 (1940). With permission.]*

- Note: The magnitude of the individual coefficients will depend on the nature of the heat transfer process (conduction, convection, condensation, boiling or radiation), on the physical properties of the fluids, on the fluid flow-rates, and on the physical arrangement of the heat-transfer surface.
- Typical values of the overall heat-transfer coefficient for various types of heat exchanger are given in Table 2.2.1.
- Fouling factors (dirt factors): Most process and service fluids will foul the heat-transfer surfaces in an exchanger to a greater or lesser extent. The deposited material will normally have a relatively low thermal conductivity and will reduce the overall coefficient.
- It is therefore necessary to oversize an exchanger to allow for the reduction in performance during operation. The effect of fouling is allowed for in design by including the inside and outside fouling coefficients.
- Fouling factors are usually quoted as heat-transfer resistances, rather than coefficients. They are difficult to predict and are usually based on past experience. Typical values for the fouling coefficients and factors for common process and service fluids are given in Table 2.2.2.

Table 2.2.1: Typical overall heat transfer coefficients

Shell and tube exchangers

Hot fluid	Cold fluid	U (W/m^2°C)
Heat exchangers		
Water	Water	800–1500
Organic solvents	Organic solvents	100–300
Light oils	Light oils	100–400
Heavy oils	Heavy oils	50–300
Gases	Gases	10–50
Coolers		
Organic solvents	Water	250–750
Light oils	Water	350–900
Heavy oils	Water	60–300
Gases	Water	20–300
Organic solvents	Brine	150–500
Water	Brine	600–1200
Gases	Brine	15–250
Heaters		
Steam	Water	1500–4000
Steam	Organic solvents	500–1000
Steam	Light oils	300–900
Steam	Heavy oils	60–450
Steam	Gases	30–300
Dowtherm	Heavy oils	50–300
Dowtherm	Gases	20–200
Flue gases	Steam	30–100
Flue	Hydrocarbon vapours	30–100
Condensers		
Aqueous vapours	Water	1000–1500
Organic vapours	Water	700–1000
Organics (some non-condensables)	Water	500–700
Vacuum condensers	Water	200–500
Vaporisers		
Steam	Aqueous solutions	1000–1500
Steam	Light organics	900–1200
Steam	Heavy organics	600–900

Table 2.2.2: Typical Fouling factors (coefficients)

Fluid	Coefficient ($W/m^2°C$)	Factor (resistance) ($m^2°C/W$)
River water	3000–12,000	0.0003–0.0001
Sea water	1000–3000	0.001–0.0003
Cooling water (towers)	3000–6000	0.0003–0.00017
Towns water (soft)	3000–5000	0.0003–0.0002
Towns water (hard)	1000–2000	0.001–0.0005
Steam condensate	1500–5000	0.00067–0.0002
Steam (oil free)	4000–10,000	0.0025–0.0001
Steam (oil traces)	2000–5000	0.0005–0.0002
Refrigerated brine	3000–5000	0.0003–0.0002
Air and industrial gases	5000–10,000	0.0002–0.0001
Flue gases	2000–5000	0.0005–0.0002
Organic vapours	5000	0.0002
Organic liquids	5000	0.0002
Light hydrocarbons	5000	0.0002
Heavy hydrocarbons	2000	0.0005
Boiling organics	2500	0.0004
Condensing organics	5000	0.0002
Heat transfer fluids	5000	0.0002
Aqueous salt solutions	3000–5000	0.0003–0.0002

- These values are for shell and tube exchangers with plain (not finned) tubes.
- The selection of the design fouling coefficient will often be an economic decision. The optimum design will be obtained by balancing the extra capital cost of a larger exchanger against the savings in operating cost obtained from the longer operating time between cleaning that the larger area will give.

Shell and tube exchangers: construction Details

- The shell and tube exchanger is by far the most commonly used type of heat-transfer equipment used in the chemical and allied industries. The advantages of this type are:
  - The configuration gives a large surface area in a small volume.
  - Good mechanical layout: a good shape for pressure operation.
  - Uses well-established fabrication techniques.
  - Can be constructed from a wide range of materials.
  - Easily cleaned.
  - Well-established design procedures.
- The principal types of shell and tube exchanger are shown in Figures 2.2.1 to 2.2.5.
- The standard nomenclature used for shell and tube exchangers is given below; the numbers refer to the features shown in Figures 2.2.1 to 2.2.5.

Nomenclature

Part number

1. Shell
2. Shell cover
3. Floating-head cover
4. Floating-tube plate
5. Clamp ring
6. Fixed-tube sheet (tube plate)
7. Channel (end-box or header)
8. Channel cover
9. Branch (nozzle)
10. Tie rod and spacer
11. Cross baffle or tube-support plate
12. Impingement baffle
13. Longitudinal baffle
14. Support bracket
15. Floating-head support
16. Weir
17. Split ring
18. Tube
19. Tube bundle
20. Pass partition
21. Floating-head gland (packed gland)
22. Floating-head gland ring
23. Vent connection
24. Drain connection
25. Test connection
26. Expansion bellows
27. Lifting ring

- The simplest and cheapest type of shell and tube exchanger is the fixed tube sheet design shown in Figure 2.2.1. The main disadvantages of this type are that the tube bundle cannot be removed for cleaning and there is no provision for differential expansion of the shell and tubes.
- The use of this type is limited to temperature differences up to about 80 °C. In the other types, only one end of the tubes is fixed and the bundle can expand freely.

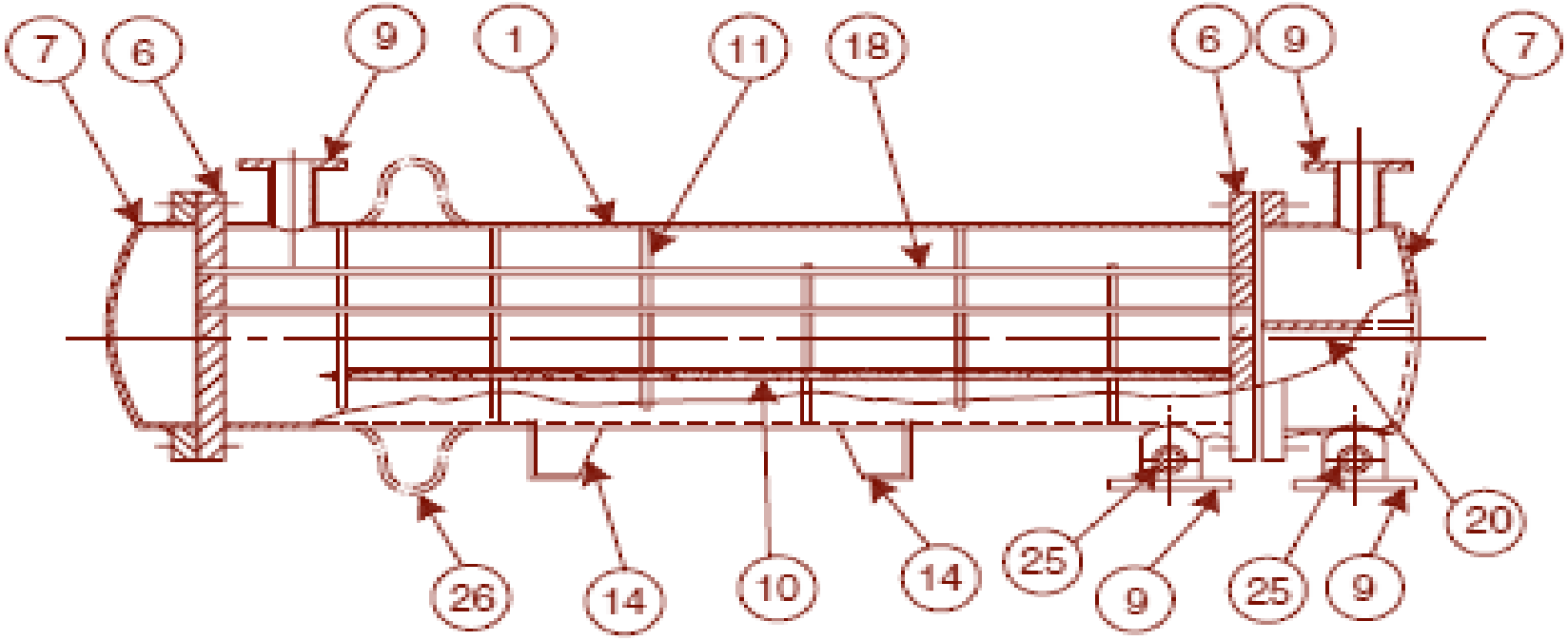


Figure 2.2.1: Fixed-tube plate (TEMA type BEM) (based on figures from BS 3274: 1960)

- The U-tube (U-bundle) type shown in Figure 2.2.2 requires only one tube sheet and is cheaper than the floating-head types; but is limited in use to relatively clean fluids as the tubes and bundle are difficult to clean. It is also more difficult to replace a tube in this type.

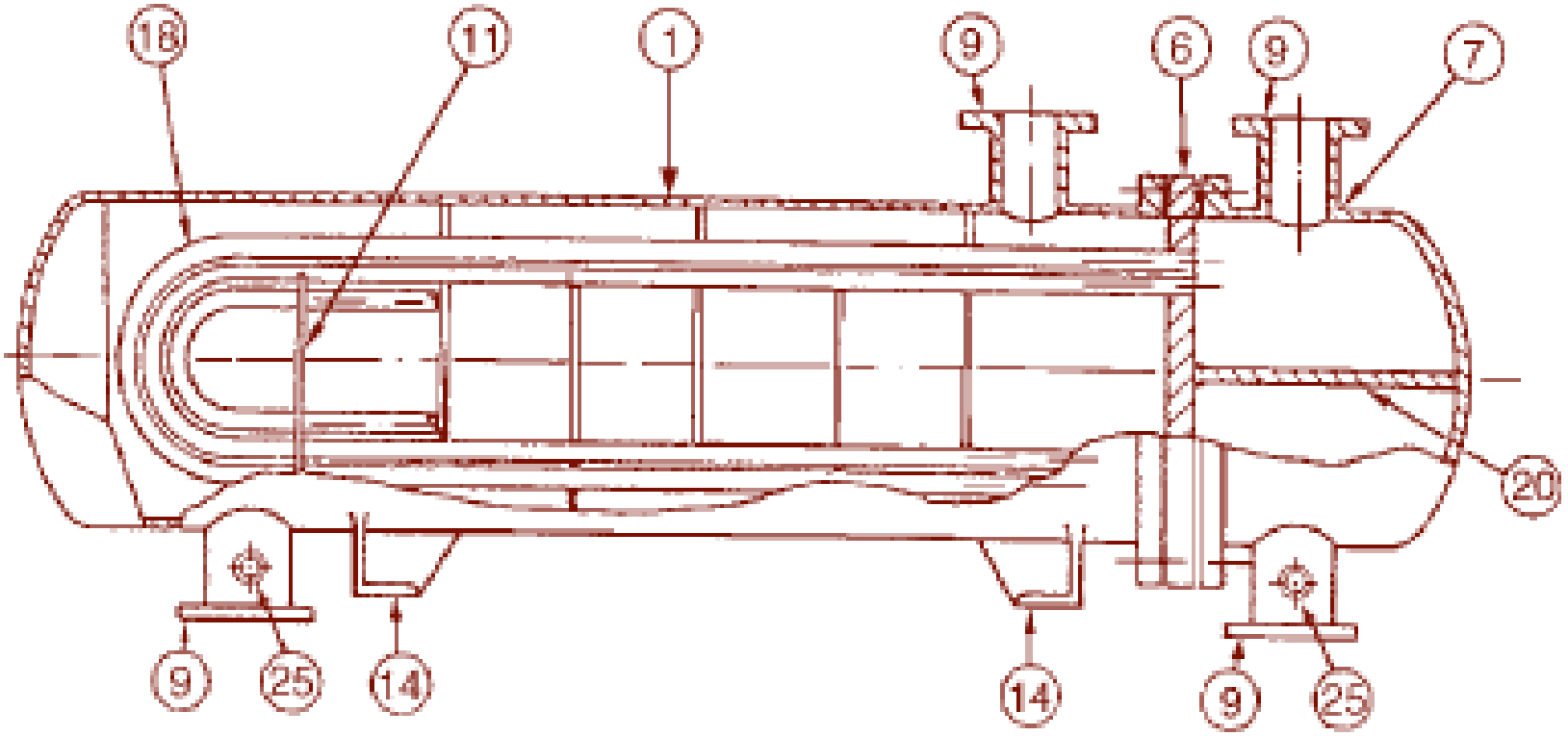


Figure 2.2.2: U-tube (TEMA type BEU) (based on figures from BS 3274: 1960)

- Exchangers with an internal floating head, Figures 2.2.3 and 2.2.4, are more versatile than fixed head and U-tube exchangers. They are suitable for high-temperature differentials and, as the tubes can be rodded from end to end and the bundle removed, are easier to clean and can be used for fouling liquids.
- A disadvantage of the pull-through design, Figure 2.2.3, is that the clearance between the outermost tubes in the bundle and the shell must be made greater than in the fixed and U-tube designs to accommodate the floating head flange, allowing fluid to bypass the tubes.
- The clamp ring (split flange design), Figure 2.2.4, is used to reduce the clearance needed. There will always be a danger of leakage occurring from the internal flanges in these floating head designs.

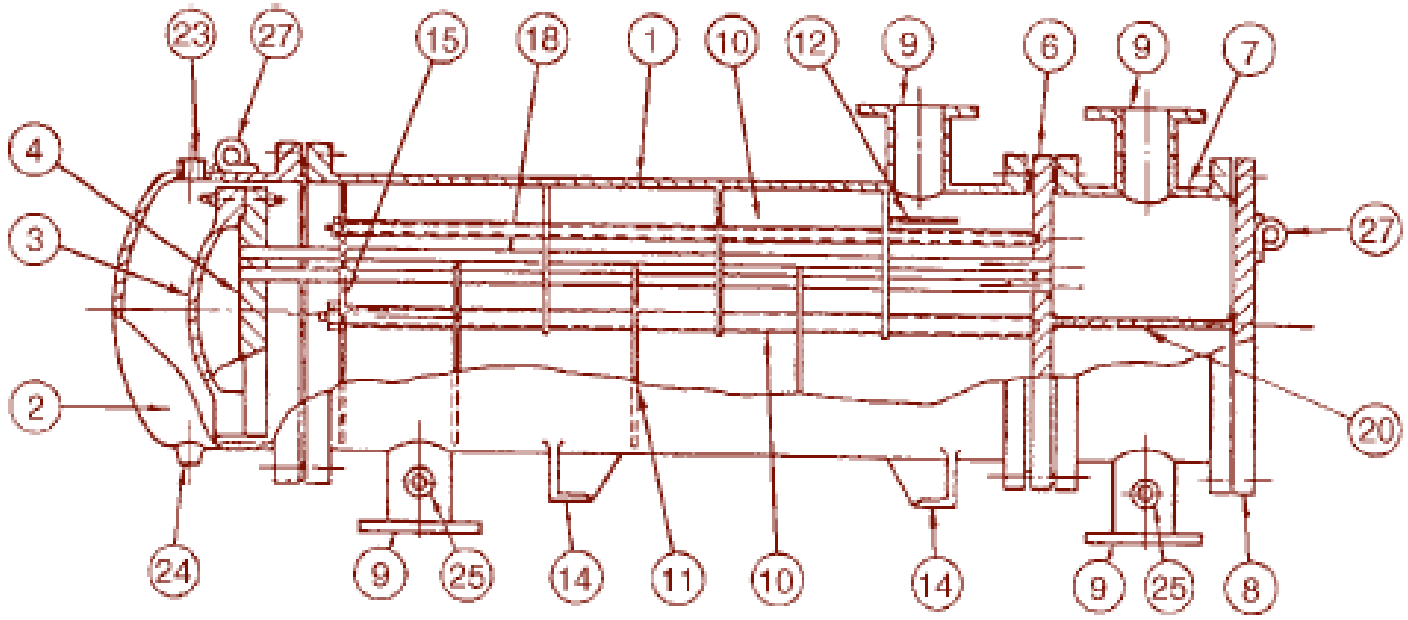


Figure 2.2.3: Internal floating head without clamp ring (TEMA type AES) (from BS 3274: 1960)

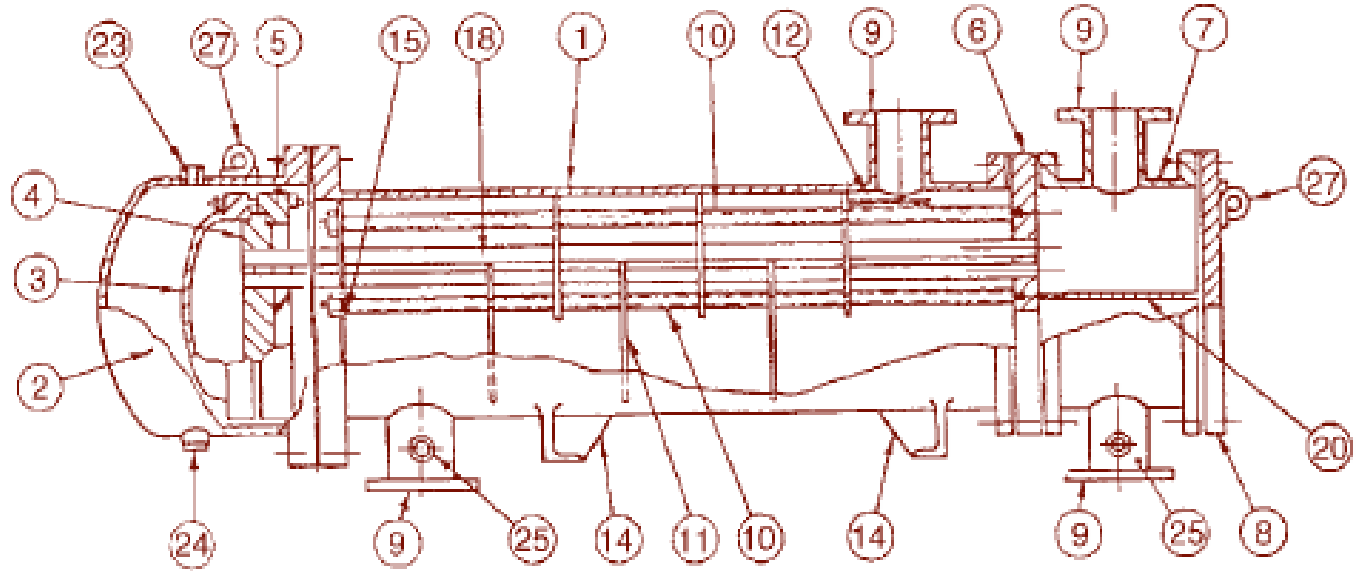


Figure 2.2.4: Internal floating head with clamp ring (TEMA type AET) (from BS 3274: 1960)

- In the external floating head designs, Figure 2.2.5, the floating-head joint is located outside the shell, and the shell sealed with a sliding gland joint employing a stuffing box. Because of the danger of leaks through the gland, the shell-side pressure in this type is usually limited to about 20 bar, and flammable or toxic materials should not be used on the shell side.

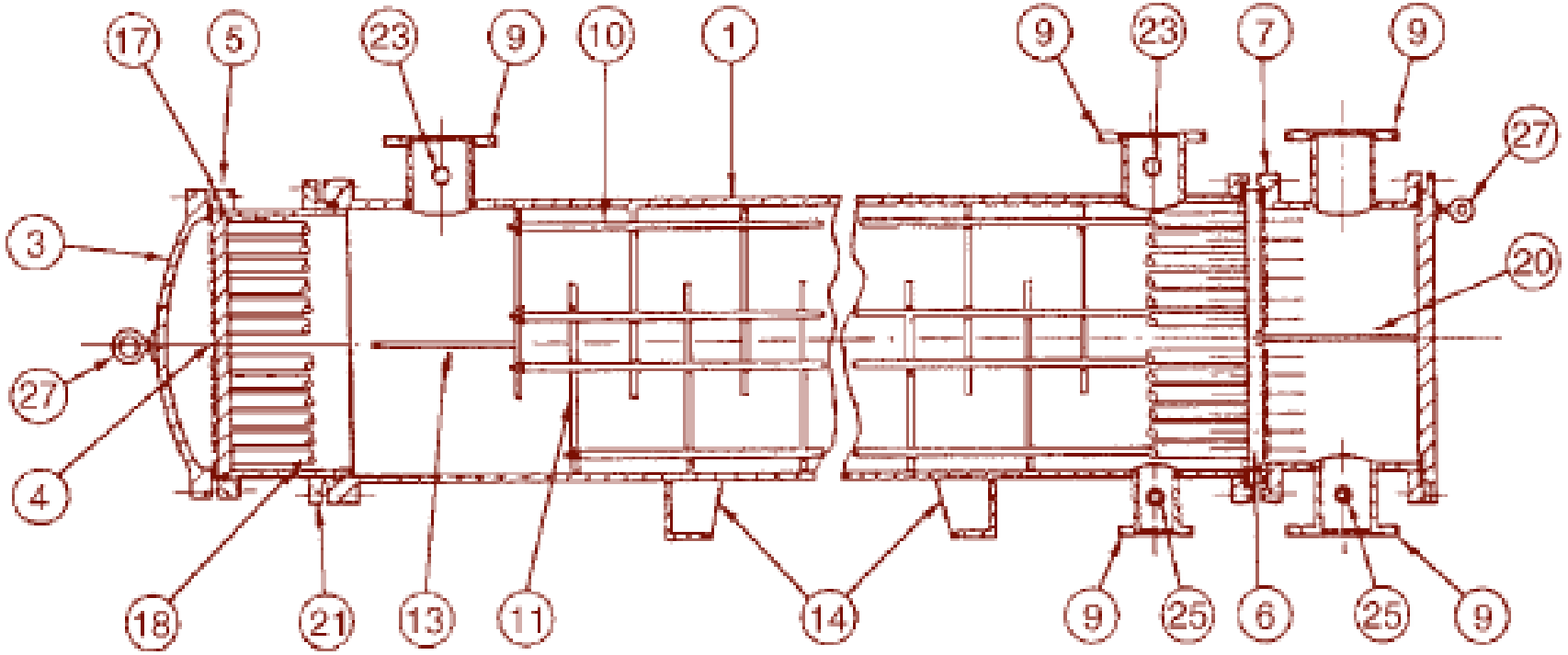


Figure 2.2.5: External floating head, packed gland (based on figures from BS 3274: 1960)

Heat-exchanger standards and codes:

- The mechanical design features, fabrication, materials of construction, and testing of shell and tube exchangers is covered by British Standard, BS 3274.
- The standards of the American Tubular Heat Exchanger Manufacturers Association, the TEMA standards, are also universally used. The TEMA standards cover three classes of exchanger: class R covers exchangers for the generally severe duties of the petroleum and related industries; class C covers exchangers for moderate duties in commercial and general process applications; and class B covers exchangers for use in the chemical process industries.
- Equivalent Indian code: IS 4503 (1967): Shell and Tube Type Heat Exchangers.
- The standards give the preferred shell and tube dimensions; the design and manufacturing tolerances; corrosion allowances; and the recommended design stresses for materials of construction.
- The shell of an exchanger is a pressure vessel and will be designed in accordance with the appropriate national pressure vessel code or standard.

Tube Dimensions:

- Tube diameters in the range 5/8 in. (16 mm) to 2 in. (50 mm) are used. The smaller diameters 5/8 to 1 in. (16 to 25 mm) are preferred for most duties, as they will give more compact, and therefore cheaper, exchangers.
- Larger tubes are easier to clean by mechanical methods and would be selected for heavily fouling fluids.
- The tube thickness (gauge) is selected to withstand the internal pressure and give an adequate corrosion allowance.
- Steel tubes for heat exchangers are covered by BS 3606 (metric sizes); the standards applicable to other materials are given in BS 3274. Standard diameters and wall thicknesses for steel tubes are given in Table 2.2.3.

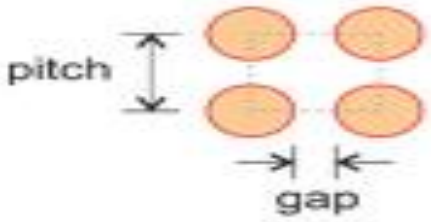
Table 2.2.3: Standard dimensions for steel tubes

Outside diameter (mm)	Wall thickness (mm)				
16	1.2	1.6	2.0	—	—
20	—	1.6	2.0	2.6	—
25	—	1.6	2.0	2.6	3.2
30	—	1.6	2.0	2.6	3.2
38	—	—	2.0	2.6	3.2
50	—	—	2.0	2.6	3.2

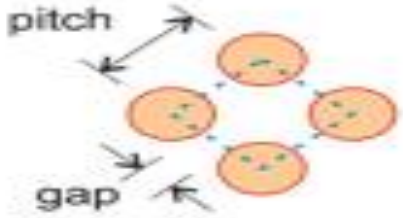
- The preferred lengths of tubes for heat exchangers are: 6 ft. (1.83 m), 8 ft (2.44 m), 12 ft (3.66 m), 16 ft (4.88 m) 20 ft (6.10 m), 24 ft (7.32 m).
- For a given surface area, the use of longer tubes will reduce the shell diameter; which will generally result in a lower cost exchanger, particularly for high shell pressures.
- The optimum tube length to shell diameter will usually fall within the range of 5 to 10.
- If U-tubes are used, the tubes on the outside of the bundle will be longer than those on the inside. The average length needs to be estimated for use in the thermal design. U-tubes will be bent from standard tube lengths and cut to size.
- The tube size is often determined by the plant maintenance department standards, as clearly it is an advantage to reduce the number of sizes that have to be held in stores for tube replacement.
- As a guide, 34 in. (19 mm) is a good trial diameter with which to start design calculations.

Tubes and tube sheets:

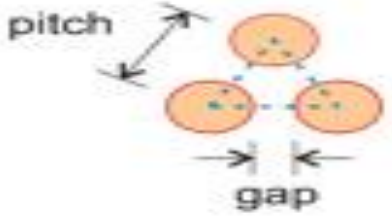
- Tubes are drawn to definite wall thickness in terms of Birmingham Wire Gauge(BWG) and true outside diameter (OD), and they are available in all common metals. Tables of dimensions of standard tubes are given in Appendix 6 (McCabe & Smith).
- Tubes are arranged on triangular or square pitch. Unless the shell side tends to foul badly, triangular pitch is used, because more heat-transfer area can be packed into a shell of given diameter than in square pitch.
- Tubes in triangular pitch cannot be cleaned by running a brush between the rows, because no space exists for cleaning lanes. Square pitch allows cleaning of the outside of the tubes. Also, square pitch gives a lower shell-side pressure drop than triangular pitch.
- TEMA standards specify a minimum center-to-center distance 1.25 times the outside diameter of the tubes for triangular pitch and a minimum cleaning lane of 0.25 in. for square pitch.



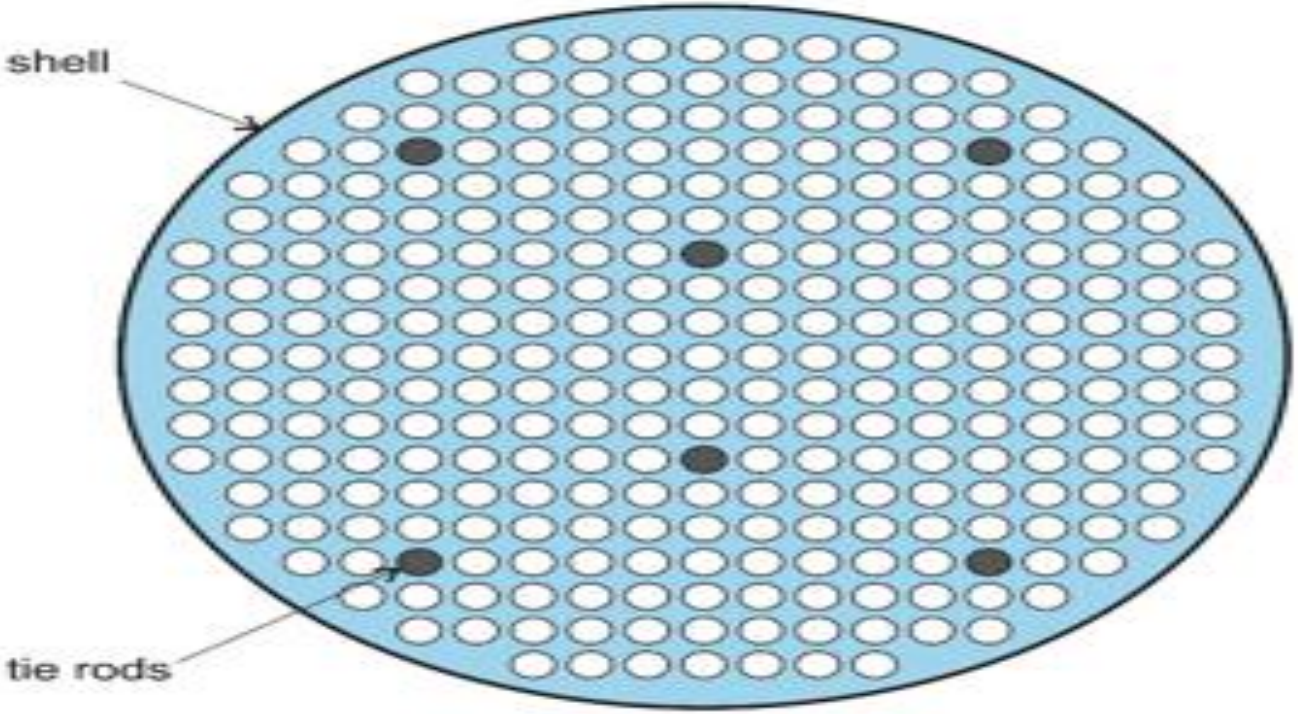
(a) square pitch



(b) rotated square pitch



(c) triangular pitch



(d) Tube bundle layout showing positions of 6 tie-rods in a bundle of 293 tubes.

Tube-sheet layout (tube count):

- The bundle diameter will depend not only on the number of tubes but also on the number of tube passes, as spaces must be left in the pattern of tubes on the tube sheet to accommodate the pass partition plates.
- An estimate of the bundle diameter D_b can be obtained from the equation given below, which is an empirical equation based on standard tube layouts. The constants for use in this equation, for triangular and square patterns, are given in Table 2.2.4.

$$D_b = d_o \left(\frac{N_t}{K_1}\right)^{1/n_1}$$

where N_t = number of tubes,
D_b = bundle diameter, mm,
d_o = tube outside diameter, mm.

Triangular pitch, $p_t = 1.25d_o$					
No. passes	1	2	4	6	8
K_1	0.319	0.249	0.175	0.0743	0.0365
n_1	2.142	2.207	2.285	2.499	2.675
Square pitch, $p_t = 1.25d_o$					
No. passes	1	2	4	6	8
K_1	0.215	0.156	0.158	0.0402	0.0331
n_1	2.207	2.291	2.263	2.617	2.643

Table 2.2.4: Constants

Shells:

- The British standard BS 3274 covers exchangers from 6 in. (150 mm) to 42 in. (1067 mm) diameter; and the TEMA standards, exchangers up to 60 in. (1520 mm).
- Up to about 24 in. (610 mm) shells are normally constructed from standard, close tolerance, pipe; above 24 in. (610 mm) they are rolled from plate.
- For pressure applications the shell thickness would be sized according to the pressure vessel design standards. The minimum allowable shell thickness is given in BS 3274 and the TEMA standards. The values, converted to SI units and rounded, are given below:

Minimum shell thickness

Nominal shell dia., mm	Carbon steel pipe	Carbon steel plate	Alloy steel
150	7.1	—	3.2
200–300	9.3	—	3.2
330–580	9.5	7.9	3.2
610–740	—	7.9	4.8
760–990	—	9.5	6.4
1010–1520	—	11.1	6.4
1550–2030	—	12.7	7.9
2050–2540	—	12.7	9.5

- In an exchanger the shell-side and tube-side heat-transfer coefficients are of comparable importance, and both must be large if a satisfactory overall coefficient is to be attained.
- The velocity and turbulence of the shell-side liquid are as important as those of the tube-side fluid. To promote crossflow and raise the average velocity of the shell-side fluid, baffles are installed in the shell.
- The baffles consist of circular disks of sheet metal with one side cut away. Common practice is to cut away a segment having a height equal to one-fourth the inside diameter of the shell. Such baffles are called 25 percent baffles. The baffles are perforated to receive the tubes. To minimize leakage, the clearances between baffles and shell and tubes should be small.
- The distance between baffles (center to center) is the baffle pitch, or baffle spacing. It should not be less than one-fifth the diameter of the shell or more than the inside diameter of the shell.



(a) Segmented baffles



(b) Double segmented baffles



(c) Triple segmented baffles



(d) Disk and donut baffles

Baffle types.

- The term "baffle cut" is used to specify the dimensions of a segmental baffle. The baffle cut is the height of the segment removed to form the baffle, expressed as a percentage of the baffle disc diameter. Baffle cuts from 15 to 45 per cent are used.
- Generally, a baffle cut of 20 to 25 per cent will be the optimum, giving good heat-transfer rates, without excessive drop.
- There will be some leakage of fluid round the baffle as a clearance must be allowed for assembly. The clearance needed will depend on the shell diameter; typical values, and tolerances, are given in Table 2.2.5.

Table 2.2.5: Typical baffle clearances and tolerances

Shell diameter, D_s	Baffle diameter	Tolerance
Pipe shells		
6 to 25 in. (152 to 635 mm)	$D_s - \frac{1}{16}$ in. (1.6 mm)	$+\frac{1}{32}$ in. (0.8 mm)
Plate shells		
6 to 25 in. (152 to 635 mm)	$D_s - \frac{1}{8}$ in. (3.2 mm)	$+0, -\frac{1}{32}$ in. (0.8 mm)
27 to 42 in. (686 to 1067 mm)	$D_s - \frac{3}{16}$ in. (4.8 mm)	$+0, -\frac{1}{16}$ in. (1.6 mm)

- Another leakage path occurs through the clearance between the tube holes in the baffle and the tubes. The maximum design clearance will normally be 1/32 in. (0.8 mm).
- The baffle spacing used range from 0.2 to 1.0 shell diameters. A close baffle spacing will give higher heat transfer coefficients but at the expense of higher pressure drop. The optimum spacing will usually be between 0.3 to 0.5 times the shell diameter.
- Where segmental baffles are used some will be fabricated with closer tolerances, 1/64 in. (0.4 mm), to act as support plates.
- The minimum spacing to be used for support plates ranges from around 1 m for 16 mm tubes to 2 m for 25 mm tubes.
- The baffles and support plate are held together with tie rods and spacers. The number of rods required will depend on the shell diameter, and will range from 4, 16 mm diameter rods, for exchangers under 380 mm diameter; to 8, 12.5 mm rods, for exchangers of 1 m diameter.

Tube sheets (plates):

- In operation the tube sheets are subjected to the differential pressure between shell and tube sides. The design of tube sheets as pressure-vessel components is covered by BS 5500.
- The joint between the tubes and tube sheet is normally made by expanding the tube by rolling with special tools.
- The tube holes are normally grooved, Figure 2.2.6a, to lock the tubes more firmly in position and to prevent the joint from being loosened by the differential expansion of the shell and tubes.
- When it is essential to guarantee a leak-proof joint the tubes can be welded to the sheet, Figure 2.2.6b. This will add to the cost of the exchanger; not only due to the cost of welding, but also because a wider tube spacing will be needed.
- The tube sheet forms the barrier between the shell and tube fluids, and where it is essential for safety or process reasons to prevent any possibility of intermixing due to leakage at the tube sheet joint, double tube-sheets can be used, with the space between the sheets vented; Figure 2.2.6c.

- To allow sufficient thickness to seal the tubes the tube sheet thickness should not be less than the tube outside diameter, up to about 25 mm diameter.
- The thickness of the tube sheet will reduce the effective length of the tube slightly, and this should be allowed for when calculating the area available for heat transfer. As a first approximation the length of the tubes can be reduced by 25 mm for each tube sheet.

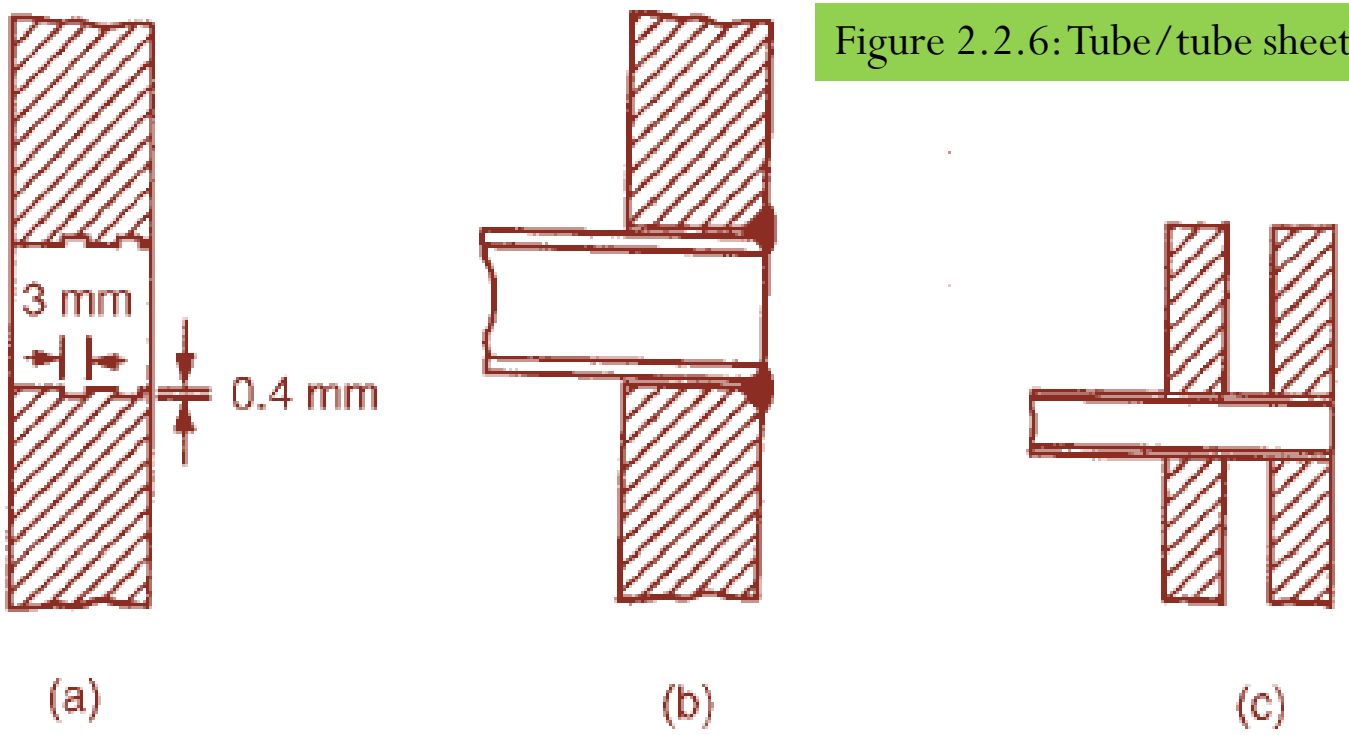


Figure 2.2.6: Tube/tube sheet joints

Fluid allocation: shell or tubes: (Where no phase change occurs)

- *Corrosion*: The more corrosive fluid should be allocated to the tube-side. This will reduce the cost of expensive alloy or clad components.
- *Fouling*: The fluid that has the greatest tendency to foul the heat-transfer surfaces should be placed in the tubes. This will give better control over the design fluid velocity, and the higher allowable velocity in the tubes will reduce fouling. Also, the tubes will be easier to clean.
- *Fluid temperatures*: If the temperatures are high enough to require the use of special alloys placing the higher temperature fluid in the tubes will reduce the overall cost. At moderate temperatures, placing the hotter fluid in the tubes will reduce the shell surface temperatures, and hence the need for lagging to reduce heat loss, or for safety reasons.
- *Operating pressures*: The higher pressure stream should be allocated to the tube-side. High-pressure tubes will be cheaper than a high-pressure shell.

- *Stream flow-rates*: Allocating the fluids with the lowest flow-rate to the shell-side will normally give the most economical design.
- *Pressure drop*: For the same pressure drop, higher heat-transfer coefficients will be obtained on the tube-side than the shell-side, and fluid with the lowest allowable pressure drop should be allocated to the tube-side.
- *Viscosity*: Generally, a higher heat-transfer coefficient will be obtained by allocating the more viscous material to the shell-side, providing the flow is turbulent. The critical Reynolds number for turbulent flow in the shell is in the region of 200. If turbulent flow cannot be achieved in the shell it is better to place the fluid in the tubes, as the tube-side heat-transfer coefficient can be predicted with more certainty.

- **Shell and tube fluid velocities**

Liquids

Tube-side, process fluids: 1 to 2 m/s, maximum 4 m/s if required to reduce fouling; water: 1.5 to 2.5 m/s.

Shell-side: 0.3 to 1 m/s.

Vapours

For vapours, the velocity used will depend on the operating pressure and fluid density; the lower values in the ranges given below will apply to high molecular weight materials.

Vacuum	50 to 70 m/s
Atmospheric pressure	10 to 30 m/s
High pressure	5 to 10 m/s

Stream temperatures:

- The closer the temperature approach used (the difference between the outlet temperature of one stream and the inlet temperature of the other stream) the larger will be the heat-transfer area required for a given duty.
- As a general guide the greater temperature difference should be at least 20 °C, and the least temperature difference 5 to 7 °C for coolers using cooling water, and 3 to 5 °C using refrigerated brines.
- Care should be taken to ensure that cooling media temperatures are kept well above the freezing point of the process materials.

Pressure drop:

Liquids

Viscosity	<1 mN s/m^2	35 kN/m^2
	1 to 10 mN s/m^2	50–70 kN/m^2

Gas and vapours

High vacuum	0.4–0.8 kN/m^2
Medium vacuum	0.1 × absolute pressure
1 to 2 bar	0.5 × system gauge pressure
Above 10 bar	0.1 × system gauge pressure

Design: Kern's method:

- This method was based on experimental work on commercial exchangers with standard tolerances and will give a reasonably satisfactory prediction of the heat-transfer coefficient for standard designs.
- The prediction of pressure drop is less satisfactory, as pressure drop is more affected by leakage and bypassing than heat transfer.
- The shell-side heat transfer and friction factors are correlated in a similar manner to those for tube-side flow by using a hypothetical shell velocity and shell diameter.
- As the cross-sectional area for flow will vary across the shell diameter, the linear and mass velocities are based on the maximum area for cross-flow: that at the shell equator.
- The shell equivalent diameter is calculated using the flow area between the tubes taken in the axial direction (parallel to the tubes) and the wetted perimeter of the tubes.

- The procedure for calculating the shell-side heat-transfer coefficient and pressure drop for a single shell pass exchanger is given below:

1. Calculate the area for cross-flow A_s for the hypothetical row of tubes at the shell equator, given by:

$$A_s = \frac{(p_t - d_o)D_s l_B}{p_t}$$

where p_t = tube pitch,
d_o = tube outside diameter,
D_s = shell inside diameter, m,
l_B = baffle spacing, m.

2. Calculate the shell-side mass velocity G_s and the linear velocity u_s:

$$G_s = \frac{W_s}{A_s}$$

$$u_s = \frac{G_s}{\rho}$$

where W_s = fluid flow-rate on the shell-side, kg/s,
ρ = shell-side fluid density, kg/m^3.

3. Calculate the shell-side equivalent diameter (hydraulic diameter).

For a square pitch arrangement:

$$d_e = \frac{4\left(p_t^2 - \frac{\pi d_o^2}{4}\right)}{\pi d_o} = \frac{1.27}{d_o}(p_t^2 - 0.785 d_o^2)$$

For an equilateral triangular pitch arrangement:

$$d_e = \frac{4\left(\frac{p_t}{2} \times 0.87 p_t - \frac{1}{2}\pi\frac{d_o^2}{4}\right)}{\frac{\pi d_o}{2}} = \frac{1.10}{d_o}(p_t^2 - 0.917 d_o^2)$$

where d_e = equivalent diameter, m.

4. Calculate the shell-side Reynolds number, given by:

$$Re = \frac{G_s d_e}{\mu} = \frac{u_s d_e \rho}{\mu}$$

5. For the calculated Reynolds number, read the value of j_h from the data plots for the selected baffle cut and tube arrangement, and calculate the shell-side heat transfer coefficient h_s from:

$$Nu = \frac{h_s d_e}{k_f} = j_h RePr^{1/3} \left(\frac{\mu}{\mu_w}\right)^{0.14}$$

Note: The tube wall temperature has to be estimated.

- For the calculated shell-side Reynolds number, read the friction factor (j_f) from data plots and calculate the shell-side pressure drop from:

$$\Delta P_s = 8 j_f \left(\frac{D_s}{d_e}\right) \left(\frac{L}{l_B}\right) \frac{\rho u_s^2}{2} \left(\frac{\mu}{\mu_w}\right)^{-0.14}$$

where L= tube length, and l_B = baffle spacing.

- The term (L/l_B) is the number of times the flow crosses the tube bundle

  $= N_b + 1$, where N_b is the number of baffles.

Tube-side heat-transfer coefficient and Pressure drop (single phase):

- Heat-transfer data for turbulent flow inside conduits of uniform cross-section are usually correlated by an equation of the form:

$$Nu = CRe^a Pr^b \left(\frac{\mu}{\mu_w}\right)^c$$

where Nu = Nusselt number = $(h_i d_e/k_f)$,
Re = Reynolds number = $(\rho u_t d_e/\mu) = (G_t d_e/\mu)$,
Pr = Prandtl number = $(C_p \mu/k_f)$
and: h_i = inside coefficient, W/m^2°C,
d_e = equivalent (or hydraulic mean) diameter, m

$$d_e = \frac{4 \times \text{cross-sectional area for flow}}{\text{wetted perimeter}} = d_i \text{ for tubes,}$$

u_t = fluid velocity, m/s,
k_f = fluid thermal conductivity, W/m°C,
G_t = mass velocity, mass flow per unit area, kg/m^{2}s,
μ = fluid viscosity at the bulk fluid temperature, Ns/m^2,
μ_w = fluid viscosity at the wall,
C_p = fluid specific heat, heat capacity, J/kg°C.

- The index for the Reynolds number is generally taken as 0.8. That for the Prandtl number can range from 0.3 for cooling to 0.4 for heating. The index for the viscosity factor is normally taken as 0.14 for flow in tubes.
- where C = 0.021 for gases,

  = 0.023 for non-viscous liquids

  =0.027 for viscous liquids
- **Laminar flow:**

$$Nu = 1.86(RePr)^{0.33}\left(\frac{d_e}{L}\right)^{0.33}\left(\frac{\mu}{\mu_w}\right)^{0.14}$$

- **Viscosity correction factor:** To apply the correction an estimate of the wall temperature is needed. This can be made by first calculating the coefficient without the correction and using the following relationship to estimate the wall temperature:

$$h_i(t_w - t) = U(T - t)$$

where t = tube-side bulk temperature (mean),
t_w = estimated wall temperature,
T = shell-side bulk temperature (mean).

- **Tube-side pressure drop:**

$$\Delta P_t = N_p \left[8 j_f \left(\frac{L}{d_i} \right) \left(\frac{\mu}{\mu_w} \right)^{-m} + 2.5 \right] \frac{\rho u_t^2}{2}$$

where j_f = dimensionless friction factor,

ΔP_t = tube-side pressure drop, N/m^2 (Pa),

N_p = number of tube-side passes,

u_t = tube-side velocity, m/s,

L = length of one tube,

m = 0.25 for laminar flow, $Re < 2100$,

= 0.14 for turbulent flow, $Re > 2100$.

Note: Another source of pressure drop will be the flow expansion and contraction at the exchanger inlet and outlet nozzles. This can be estimated by adding one velocity head for the inlet and 0.5 for the outlet, based on the nozzle velocities.

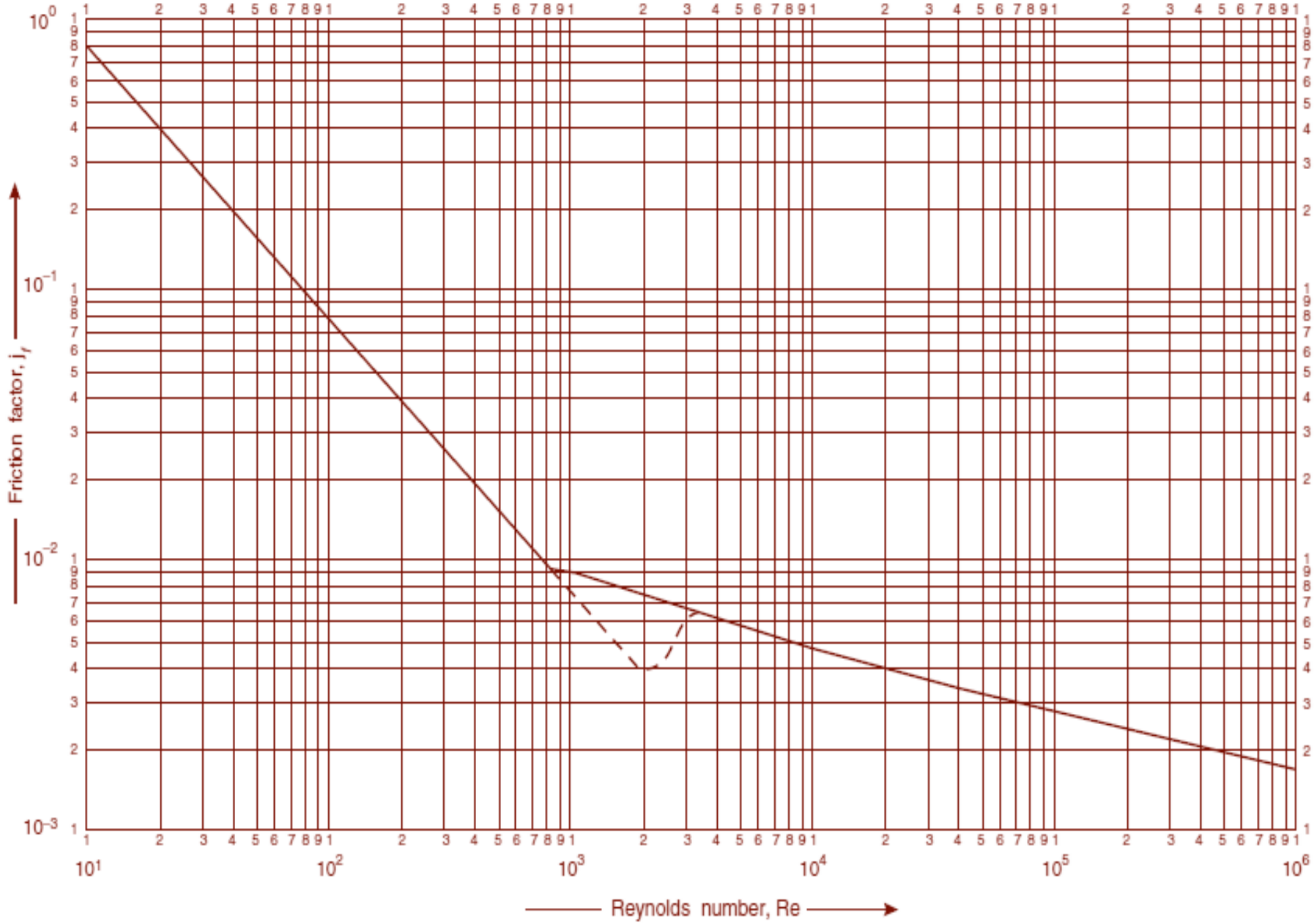


Figure 12.24. Tube-side friction factors

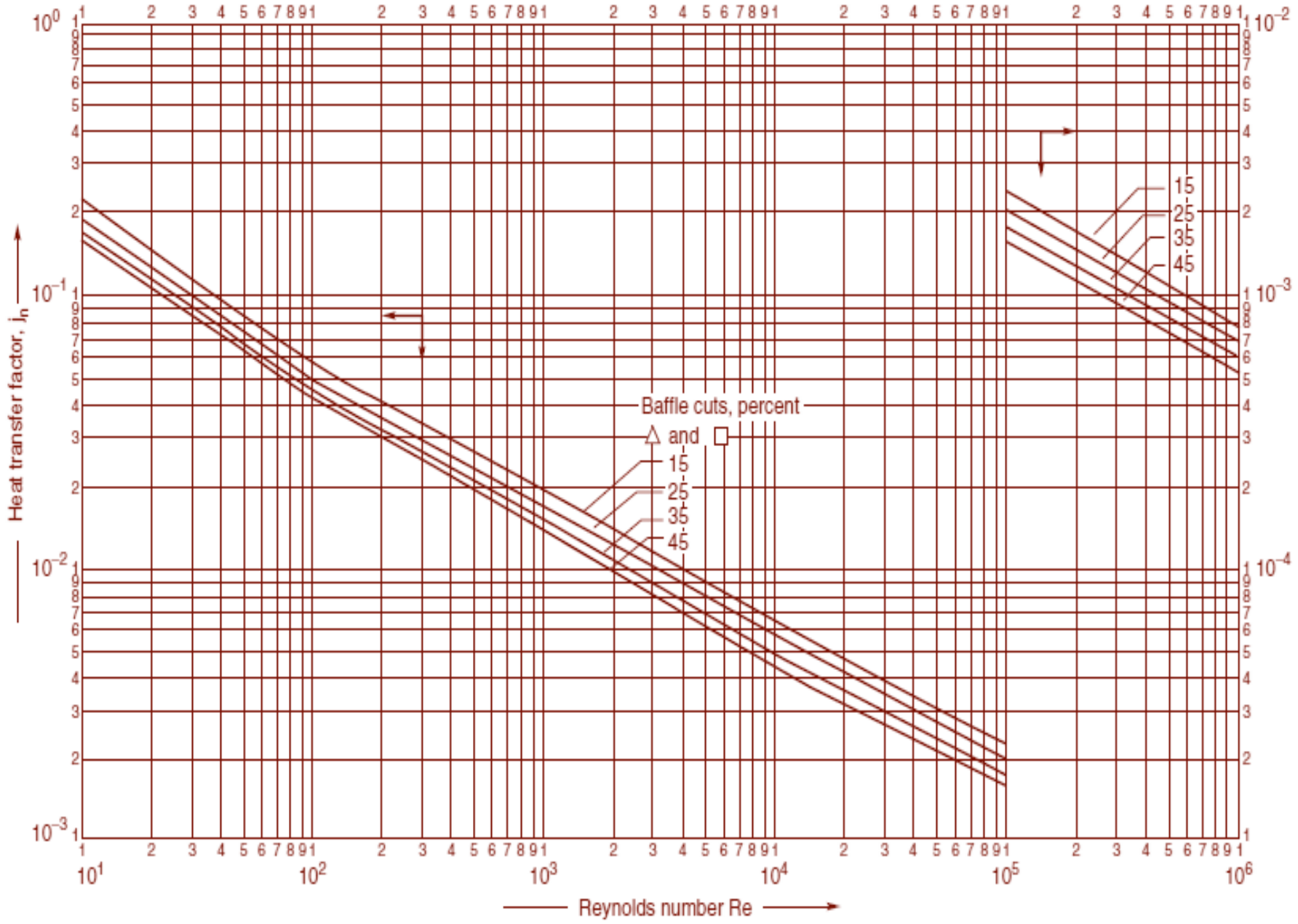
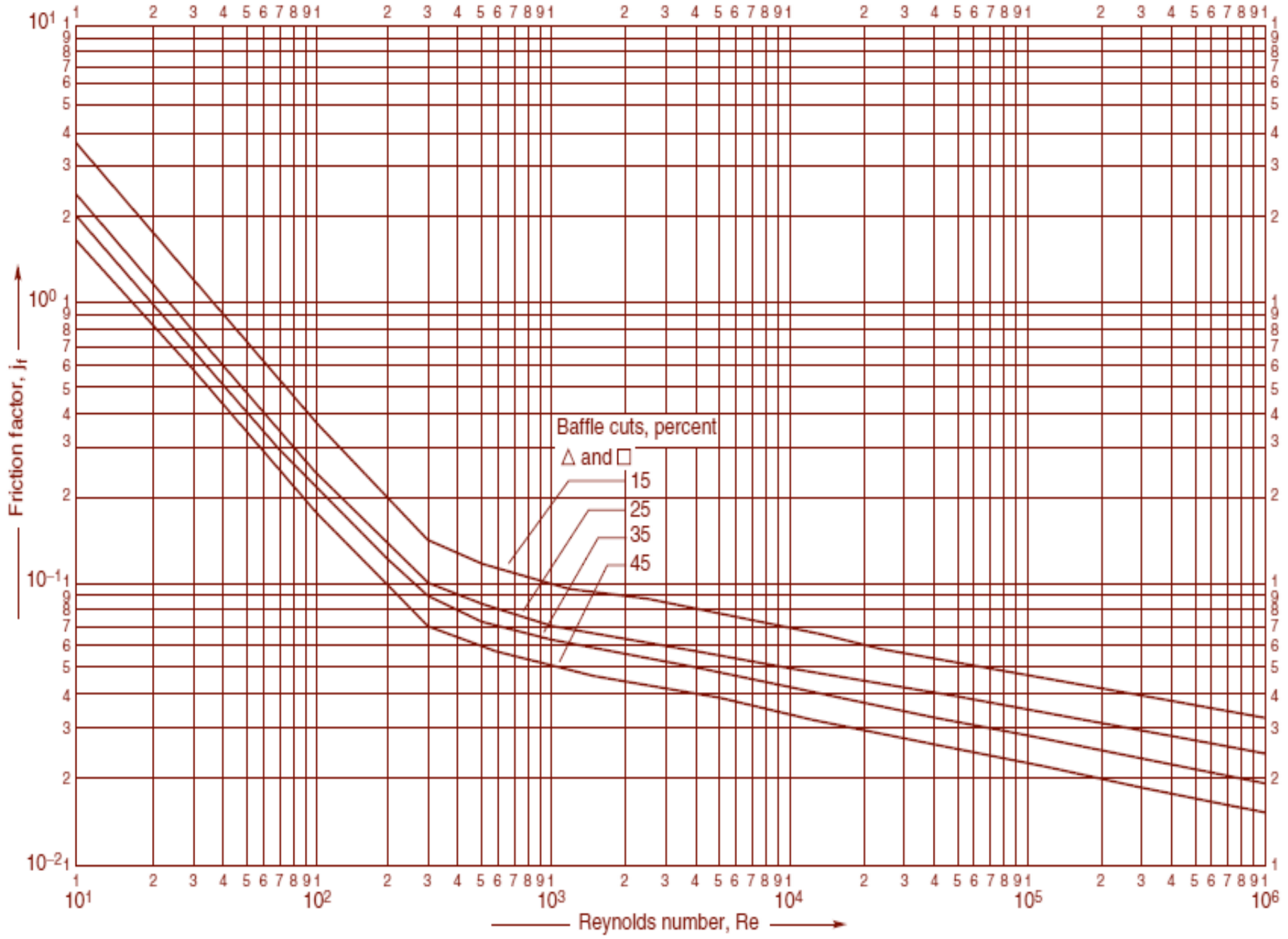


Figure 12.29. Shell-side heat-transfer factors, segmental baffles


Friction factor, j_f
Reynolds number, Re
Baffle cuts, percent
△ and □
15
25
35
45

Example-1:

Design an exchanger to sub-cool condensate from a methanol condenser from 95 °C to 40 °C. Flow-rate of methanol 100,000 kg/h. Brackish water will be used as the coolant, with a temperature rise from 25 °C to 40 °C.

Solution: Coolant is corrosive, so assign to tube-side.

Step 1: Get the heat capacity data of methanol and water (from data). Calculate the heat duty of methanol and cooling water flowrate from the energy balances. Calculate the LMTD.

Step 2: Assume type of heat exchanger (Ex: one shell pass and two tube passes), Calculate the correction factor (from data) for LMTD and corrected LMTD.

Step 3: Assume overall heat transfer coefficient (from data) and calculate the provisional area.

Step 4: Choose tube dimensions (from standard tube tables), calculate number of tubes, bundle diameter, and shell diameter.

Choose 20 mm o.d., 16 mm i.d., 4.88-m-long tubes $\left(\frac{3}{4}\text{in.} \times 16\text{ ft}\right)$, cupro-nickel.

Allowing for tube-sheet thickness, take

$$L = 4.83 \text{ m}$$

$$\text{Area of one tube} = 4.83 \times 20 \times 10^{-3}\pi = 0.303 \text{ m}^2$$

$$\text{Number of tubes} = \frac{278}{0.303} = \underline{\underline{918}}$$

As the shell-side fluid is relatively clean use 1.25 triangular pitch.

$$\text{Bundle diameter } D_b = 20\left(\frac{918}{0.249}\right)^{1/2.207} = 826 \text{ mm}$$

Use a split-ring floating head type.

From Figure 12.10, bundle diametrical clearance = 68 mm,

$$\text{shell diameter, } D_s = 826 + 68 = 894 \text{ mm.}$$

(*Note*. nearest standard pipe sizes are 863.6 or 914.4 mm).

Shell size could be read from standard tube count tables.

Step 5: Calculate tube side coefficient.

$$\text{Mean water temperature} = \frac{40 + 25}{2} = 33^{\circ}\text{C}$$

$$\text{Tube cross-sectional area} = \frac{\pi}{4} \times 16^2 = 201 \text{ mm}^2$$

$$\text{Tubes per pass} = \frac{918}{2} = 459$$

$$\text{Total flow area} = 459 \times 201 \times 10^{-6} = 0.092 \text{ m}^2$$

$$\text{Water mass velocity} = \frac{68.9}{0.092} = 749 \text{ kg/s m}^2$$

$$\text{Density water} = 995 \text{ kg/m}^3$$

$$\text{Water linear velocity} = \frac{749}{995} = 0.75 \text{ m/s}$$

$$\frac{h_i d_i}{k_f} = j_h Re Pr^{0.33} \left(\frac{\mu}{\mu_w}\right)^{0.14}$$

Viscosity of water = 0.8 mNs/m^2

Thermal conductivity = 0.59 W/m°C

$$Re = \frac{\rho u d_i}{\mu} = \frac{995 \times 0.75 \times 16 \times 10^{-3}}{0.8 \times 10^{-3}} = 14{,}925$$

$$Pr = \frac{C_p \mu}{k_f} = \frac{4.2 \times 10^3 \times 0.8 \times 10^{-3}}{0.59} = 5.7$$

Neglect $\left(\frac{\mu}{\mu_w}\right)$

$$\frac{L}{d_i} = \frac{4.83 \times 10^3}{16} = 302$$

From Figure 12.23, $j_h = 3.9 \times 10^{-3}$

$$h_i = \frac{0.59}{16 \times 10^{-3}} \times 3.9 \times 10^{-3} \times 14{,}925 \times 5.7^{0.33} = 3812 \text{ W/m}^2\text{°C}$$

Step 6: Calculate the shell side coefficient.

Choose baffle spacing $= \frac{D_s}{5} = \frac{894}{5} = 178$ mm.

$$\text{Tube pitch} = 1.25 \times 20 = 25 \text{ mm}$$

$$\text{Cross-flow area } A_s = \frac{(25-20)}{25} 894 \times 178 \times 10^{-6} = 0.032 \text{ m}^2$$

$$\text{Mass velocity, } G_S = \frac{100{,}000}{3600} \times \frac{1}{0.032} = 868 \text{ kg/s m}^2$$

$$\text{Equivalent diameter } d_e = \frac{1.1}{20}(25^2 - 0.917 \times 20^2) = 14.4 \text{ mm}$$

$$\text{Mean shell side temperature} = \frac{95+40}{2} = 68°\text{C}$$

Methanol density $= 750$ kg/m^3

Viscosity $= 0.34$ mNs/m^2

Heat capacity $= 2.84$ kJ/kg°C

Thermal conductivity $= 0.19$ W/m°C

$$Re = \frac{G_s d_e}{\mu} = \frac{868 \times 14.4 \times 10^{-3}}{0.34 \times 10^{-3}} = 36{,}762$$

$$Pr = \frac{C_p \mu}{k_f} = \frac{2.84 \times 10^3 \times 0.34 \times 10^{-3}}{0.19} = 5.1$$

Choose 25 per cent baffle cut, from Figure 12.29

$$j_h = 3.3 \times 10^{-3}$$

Without the viscosity correction term

$$h_s = \frac{0.19}{14.4 \times 10^{-3}} \times 3.3 \times 10^{-3} \times 36{,}762 \times 5.1^{1/3} = 2740 \text{ W/m}^2\text{°C}$$

Estimate wall temperature

$$\text{Mean temperature difference} = 68 - 33 = 35°\text{C}$$

across all resistances

$$\text{across methanol film} = \frac{U}{h_o} \times \Delta T = \frac{600}{2740} \times 35 = 8°\text{C}$$

$$\text{Mean wall temperature} = 68 - 8 = 60°\text{C}$$

$$\mu_w = 0.37 \text{ mNs/m}^2$$

$$\left(\frac{\mu}{\mu_w}\right)^{0.14} = 0.99$$

which shows that the correction for a low-viscosity fluid is not significant.

Step 7:Take thermal conductivity data and fouling coefficient values and calculate the overall coefficient.

Step 8: Calculate the tube side pressure drop and compare with the maximum allowable pressure drop, and can increase the number of passes if it is low.

Step 9: Calculate shell side pressure drop and reduce it by increasing the baffle pitch, if it is more than the allowable limit. Revise the coefficients.

Bell's method:

- In Bell's method the heat-transfer coefficient and pressure drop are estimated from correlations for flow over ideal tube-banks, and the effects of leakage, bypassing and flow in the window zone are allowed for by applying correction factors.
- This approach will give more satisfactory predictions of the heat-transfer coefficient and pressure drop than Kern's method; and, as it takes into account the effects of leakage and bypassing, can be used to investigate the effects of constructional tolerances and the use of sealing strips.
- The shell-side heat transfer coefficient is given by:

$$h_s = h_{oc} F_n F_w F_b F_L$$

where h_{oc} = heat transfer coefficient calculated for cross-flow over an ideal tube bank, no leakage or bypassing.

F_n = correction factor to allow for the effect of the number of vertical tube rows,

F_w = window effect correction factor,

F_b = bypass stream correction factor,

F_L = leakage correction factor.

The total correction will vary from 0.6 for a poorly designed exchanger with large clearances to 0.9 for a well-designed exchanger.

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